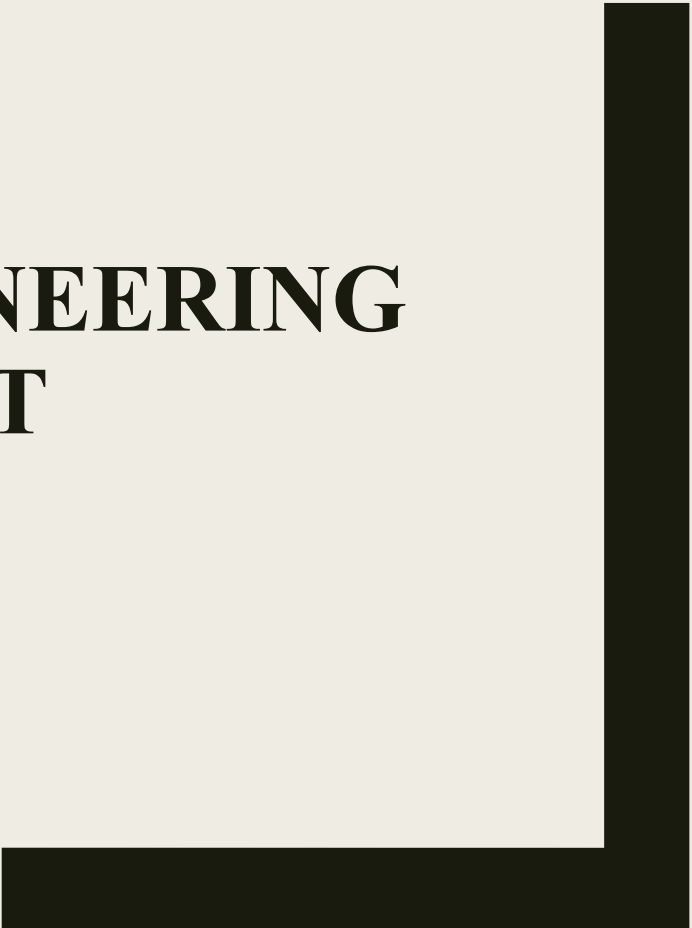




INDUSTRIAL WASTE ENGINEERING AND MANAGEMENT



MODULE II

Disposal of treated waste into rivers. De-oxygenation and Re- oxygenation in river and Streams, Development of Oxygen sag model, Application of Streeter Phelp's equation, Ocean disposal, Water reclamation and reuse.

References:

1. Metcalf and Eddy. *Wastewater Engineering, Treatment, Disposal and Reuse*. Tata McGraw Hill Publications, 1991.
2. Nemerow, H. N. *Liquid Waste from Industry – Theory, Practice and Treatment*. Addison-wesley, 1971.
3. Rao, M.N. and Datta, A.K. *Waste Water Treatment*. Oxford IBH Publication, 2017.
4. Eckenfelder, W.W. *Industrial Water Pollution Control*. McGraw Hill Publication, 1999.
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Insights

1. Ganga, Prayagraj

The river Ganga occupies a unique position in Hindu culture and is considered the holiest river in India. The river flows through 29 Indian cities with huge populations which means there is a proportional increase in human, Industrial waste and religious events.



2. Yamuna, Delhi

River Yamuna is one of the most polluted rivers of the India and the second largest tributary river of the Ganga. The holy river is getting polluted especially around New Delhi with garbage and industrial sewage.



3. Sabarmati, Ahmedabad

Sabarmati River is one of the major west flowing river of Gujarat and third most polluted river in the country. It originates from Aravali hill ranges in Rajasthan and flows towards the Gulf of Cambay in Gujarat.



4. Brahmaputra, Assam

The Brahmaputra is facing the crisis of pollution in Assam. The 2,900 km long river, the lifeline of the region is in an extremely hazardous condition now. Suddenly, the water of the Brahmaputra River in Assam has turned muddy and has changed its colour.



5. Cooum, Chennai

Cooum river flows through the major corporation zones of Chennai and the river is highly polluted in this urban area. Polluted part of the Cooum river is fed by pesticides of corporation zones and heavy metals from the industry in Kilpauk and Triplicane.



6. Gomti, Lucknow

Gomti river is one of the tributary of the Ganga river and one of the many transcendental rivers in India. The river collects large amounts of human and industrial pollutants from city of Lucknow, Lakhimpur Kheri, Sitapur and Jaunpur.



Insights contd...

- <https://www.youtube.com/watch?v=eHCalvPN2rY>
- <https://www.youtube.com/watch?v=qUFqRcBfxjc>
- <https://www.youtube.com/watch?v=FTumcXtIYJM>
- <https://www.youtube.com/watch?v=Q0gYQrebGwY>



Mass fish kill was reported early on Tuesday after the suspected release of effluents from industries upriver in the Eloor-Edayar in Ernakulam district. Fishermen have reported losses amounting to lakhs of rupees and are urging authorities to find ways to look into the matter and grant cost relief.



The dead fish started floating near the bridge at Pathalam. (Photo:Jipson Sikhera).

Kerala: Periyar fish kill sparks outrage amid uncertainty over pollution factors

Public seek action, stage stir; PCB says fish died due to depletion of oxygen levels in river



The protestors threw rotting fish into the Eloor PCB office compound (Photo | Express)

Express News Service

Updated on: 23 May 2024, 10:28 am · 3 min read



KOCHI: The recent mass death of fish in the Periyar has sparked outrage among cage fish farmers and local representatives, leading to heated protests and demands for immediate action. On Wednesday, the Kerala State Pollution Control Board (KSPCB) convened a crucial meeting at its Eloor office to address the escalating situation. Panchayat presidents from affected areas, MLA T J Vinod, chief environmental engineer of the district, and officials from relevant departments were present at the meeting in which key decisions were made regarding the crisis.

Water, which has been coloured by textile dye and will eventually flow through the Labandha, Turag and Buriganga rivers, is released near a paddy field in the Mawna Union area, north of Gazipur.



Waste Water Disposal Methods

■ Introduction

- The disposal of sewage effluent is the last stage of getting rid of sewage
- After subjecting it to various steps of processes
- Treatment of transforming the sewage into a harmless liquid which fulfils the minimum standard of health and sanitation.
- The main objects of controlling the disposal of sewage are
 - • *To render the sewage inoffensive*
 - • *To save the aquatic life in streams*
 - • *To eliminate the danger of contamination of water supplies.*
- Amount or degree of treatment – the source of its disposal, capacity to assimilate the impurities present in the sewage without itself getting polluted or less useful
- Before designing the treatment plant, the source of disposal has to be selected first.

Methods of Wastewater Disposal

- 1. Natural Methods
- 2. Artificial Methods
- 3. Combined Methods
- **Natural Methods**
 - (i) *Dilution or disposal into water i.e. into sea, lakes or rivers*
 - (ii) *Disposal on land or land treatment i.e. sewage farming and irrigation*
- **Artificial Methods**
- By which the sewage is disposed off only after subjecting it to various treatments (primary and secondary) such as:
 - (i) *Screening and detritus removal*
 - (ii) *Sedimentation with or without chemicals*
 - (iii) *Biological treatment (trickling filter, oxidation pond or activated sludge process)*
- The actual practice is to use both the methods, the sewage is first given the treatment and then it is disposed off by any of the natural method. If full treatment is not given at least the primary treatment is given before disposal.

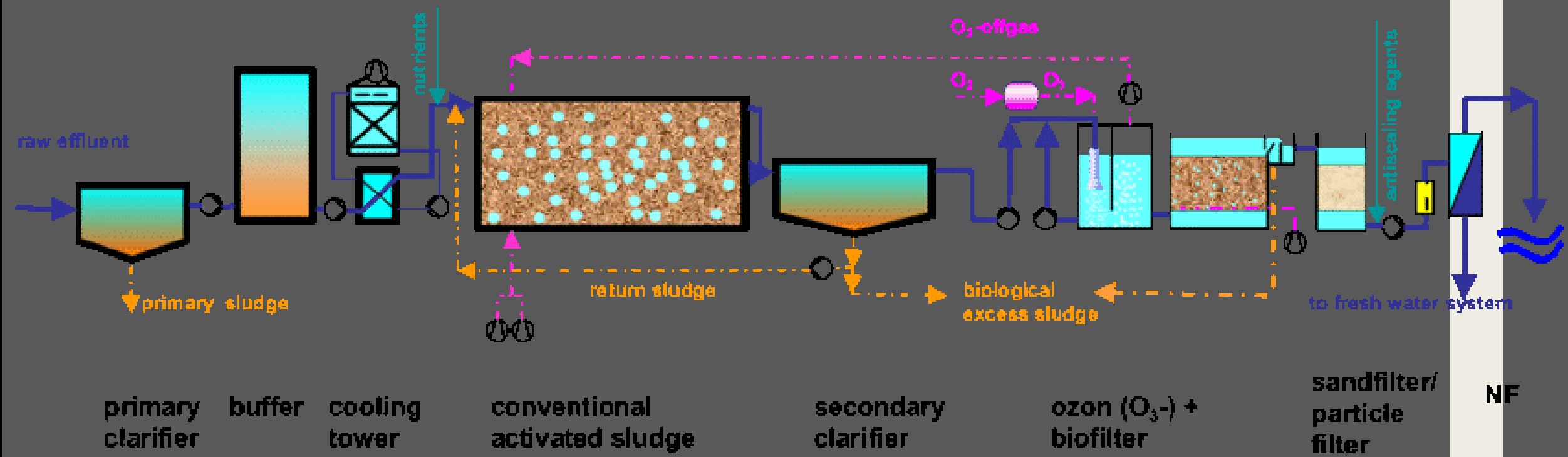
Methods of Wastewater Disposal contd...

- **Combined Methods**
- If the strength of sewage exceeds selfpurification capacity of river, then the river can't regain its original quality.
- To avoid this natural methods of disposal are combined with artificial methods of disposal.

primary treatment

secondary treatment

tertiary treatment chemical- biochemical oxidation + (partial) desalination



Natural Method

- *Dilution:*

Disposal of treated waste into river, lakes, sea etc



Natural Method contd...

■ *Dilution:*

- When sufficient dilution water is available in the receiving water body, where the wastewater is discharged, the DO level in the receiving stream may not reach to zero or critical DO due to availability of sufficient DO initially in the river water before receiving discharge of wastewater
- Receiving body - *river, sea, lake*

Natural Method contd...

■ *Dilution:*

- the depletion of oxygen would kill the aquatic life, and if this dilution water is used at the downstream side for drinking purposes, it will cause danger to the public health.
- If diluting water is not sufficient to supply the biochemical oxygen demand to oxidize the entire matter present, there will be nuisance of foul odour and unsightly islands of half digested floating, putrefying matter at the surface

Conditions Favouring Disposal

- The Dilution method for disposing of the sewage can be favourable by adopted under the following conditions:
 - • *When the sewage is comparatively fresh i.e. it is discharged within 3-4 hours of its collection.*
 - • *When the floating matter and settleable solids have been removed by primary treatment.*
 - • *When the diluting water has high DO content, so that not only the BOD is satisfied, but sufficient DO remains available for the aquatic life.*

Conditions Favouring Disposal Contd...

- • *Where current of the diluting waters flow are favourable, causing no deposition or destruction to aquatic life. It means that swift forward currents are helpful, as they easily carry away the sewage to the point of unlimited dilution. On the other hand, slow back currents tend to cause sedimentation, resulting in large sludge deposits.*
 - • *Where the wastewater does not contain industrial wastewater having toxic substances.*
 - • *When the outfall sewer of the city or the treatment plant is situated near some natural waters.*
- The bureau of Indian standards has therefore laid down standards for sewage effluents (IS 4764-1973) and for industrial effluents (IS 2490 – 1974).
- When the industrial waste water are disposed of in to public sewers, their quality should be control by using the standards IS 3306-1974.

Types of Receiving Waters for Dilution

- The following are the types of receiving waters into which wastewater or effluent can be discharged for dilution:
 - *Perennial rivers and streams*
 - *Lakes*
 - *Oceans or Sea*
 - *Estuaries*
 - *Creeks*

Perennial rivers and streams

- Best type of receiving waters, since the water is in continuous motion.
- Also in the natural streams there is a balance between plant and animal life, with considerable interaction among the various life forms.
- The discharge flowing during summer and during winter varies.
- During summer, minimum flow in the stream low dilution factor and high temperature of water may result in low solubility of oxygen, necessitating proper treatment before dilution.

Lakes

- Used for dilution when the perennial streams are not available
- Depends on characteristics of lakes, such as its size, shape, volume of fresh water flowing into it etc.

Ocean

- Ocean has abundant water --- **dilution factor is unlimited.**
- Sea water has **20 % less DO** than rivers or streams.
- The water is **turbid** due to dissolved impurities and penetration of **sun rays is less.**
- Proper care has to be taken in dilution by sea otherwise **anaerobic conditions** would occur resulting in the forming of sludge banks and emission of foul odour.

Estuary and Creek

■ Creek

- A creek is in the form of an inlet on sea coast, may not have dry weather flow during some part of the year.
- Due to this, great care should be taken in disposal of effluent in to it.

■ Estuary

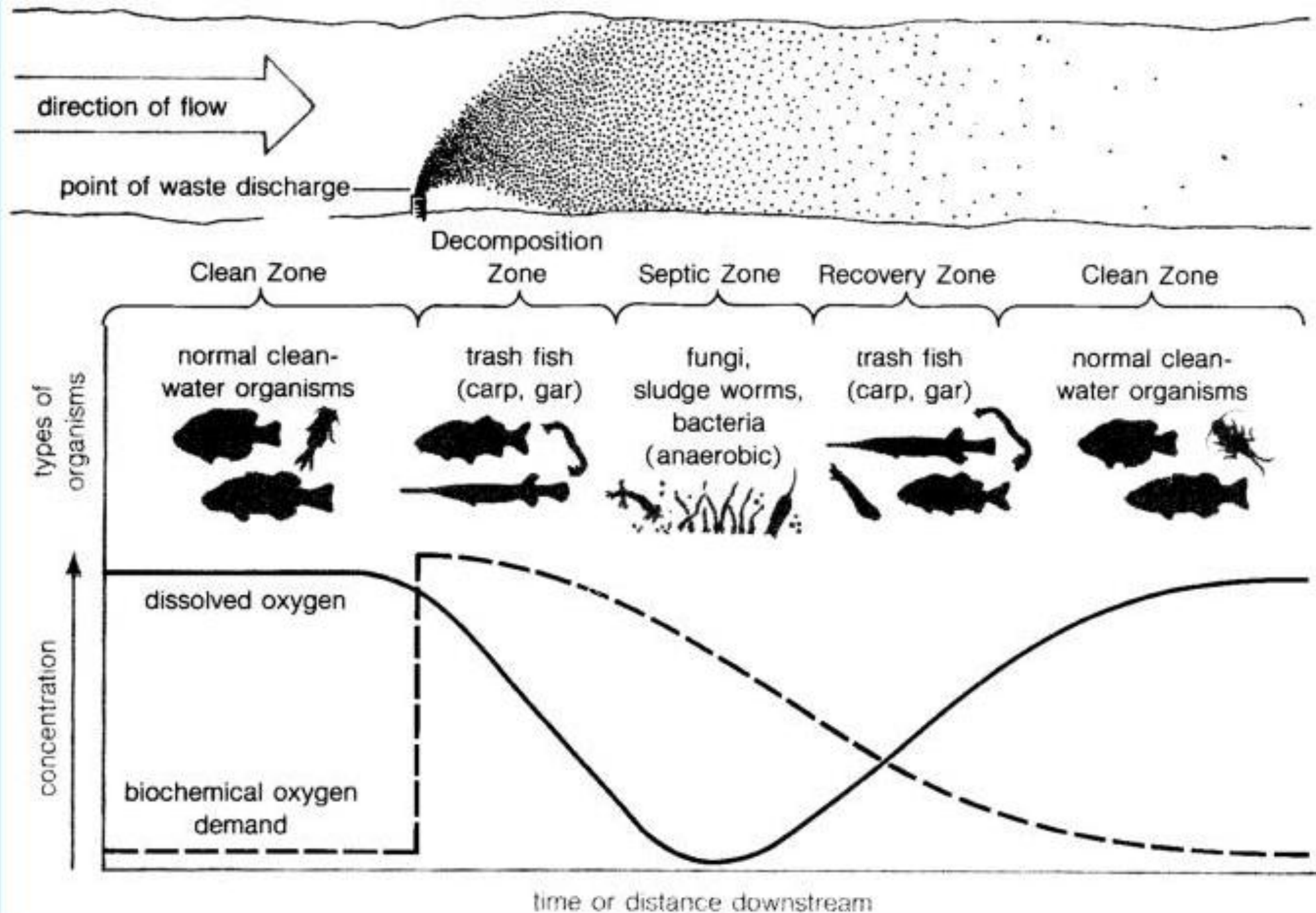
- Estuary is wide lower tidal part of the river.
- Hence dilution in an estuary is affected both by ocean water as well as river water.
- Process of dilution is generally satisfactory in estuaries.

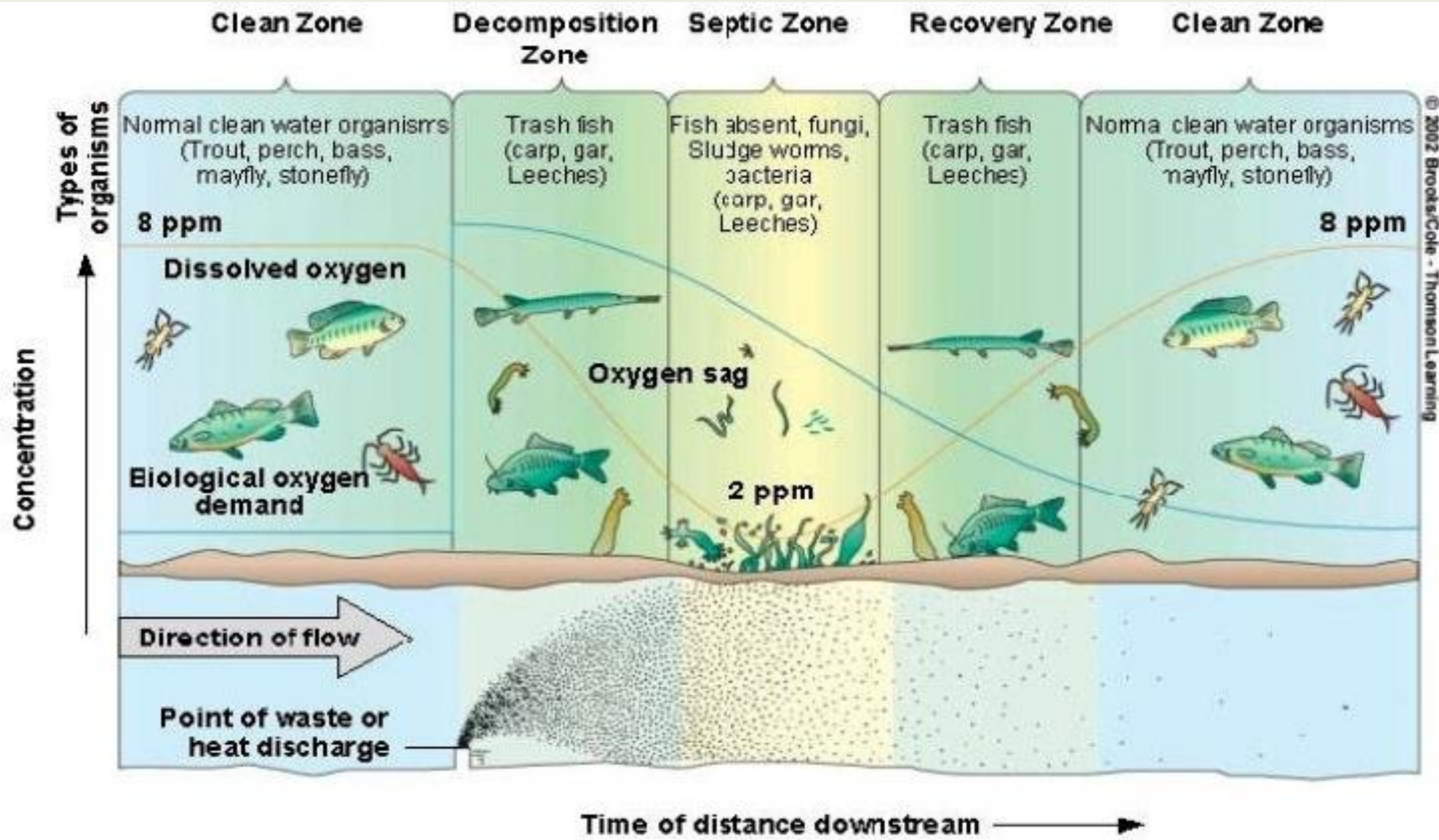
Self-purification of water bodies

- Wastewater or the effluent is discharged into a natural stream, the organic matter is broken down by bacteria to ammonia, nitrates, sulphates, carbon dioxide etc.
- In this process of oxidation, DO content of natural water is utilized.
- Creating a deficiency of DO
- As the excess organic matter is stabilized, the normal cycle will be re-established in a process known as self-purification where in the oxygen is replenished by its aeration by wind.

Self-purification of water bodies

- *Complex process involves physical, chemical and biological processes*
- *Indicator of river health– Dissolved Oxygen content-4ppm*
- *DO available in water depends on no of factors*
 - Oxygen demanding waste reduce DO
 - Plants add DO during day time but removes during night
 - Respiration of organism removes oxygen
 - Summer, temp rise reduces DO in river
 - Slow movement of water reduces DO





PARTI- CULARS	ZONES OF POLLUTION				
	CLEAR WATER	ZONE OF DEGRADATION	ZONE OF ACTIVE DECOMPOSITION	ZONE OF RECOVERY	ZONE OF CLEARER WATER
1. DO SAG CURVE	 SATURATION LEVEL 100% 40% DO ZERO 100%				
2. PHYSICAL INDICES	CLEAR WATER NO BOTTOM SLUDGES	FLOATING SOLIDS, BOTTOM SLUDGE, COLOUR, TURBIDITY	GREYISH AND DARKER COLOUR, CH_4 , H_2S , CO_2 , N_2 GASES EVOLUTION, BLACK SCUM AT TOP	TURBID WITH BOTTOM SLUDGE	CLERER WATER WITH NO BOTTOM SLUDGE

1. Zone of degradation/decomposition

- This zone is situated just below the outfall sewer when discharging its contents into the stream.
- The DO is reduced to 40% of the saturation values.
- There is an increase in carbon dioxide content, and reaeration is much slower than deoxygenation.
- Though conditions are unfavorable for aquatic life, fungi at higher points and bacteria at lower points breed small worms which work over and stabilize sewage sludge.
- The decomposition of solid matter takes place in this and anaerobic decomposition prevails.

2. Zone of active decomposition/Septic zone

- This zone is just after the degradation zone and is marked by heavy pollution.
- Water in this zone becomes greyish and darker than the previous zone.
- The DO concentration in this zone falls down to zero.
- Active anaerobic decomposition take place with the evolution of methane, hydrogen sulphide, carbon dioxide and nitrogen.
- Protozoa and fungi first disappear and then reappear, while algae will mostly be absent.
- Near the end of this zone, as the decomposition slackens, reaeration sets in and DO again rises to its original level of 40%.

3. Zone of recovery

- In this zone the process of recovery starts, from its degraded condition to its former condition.
- The stabilization of organic matter takes place in this zone.
- Due to this most of the stabilized organic matter settles as sludge, BOD falls and DO content rises above the 40% value.
- Mineralization is active, with the resulting formation of products like nitrates, sulphates and carbonates.
- Near the end of this zone, microscopic aquatic life reappears, fungi decrease and algae reappears.

4. Zone of clear water

- In this zone the natural condition of the stream is restored with the result that
 - Water becomes clear and attractive in appearance.
 - DO rises to the saturation level and is much higher than the BOD and
 - Oxygen balance is attained.
- The recovery is said to be complete in this zone, though some pathogenic organisms present in this zone.
- Disposal of Wastewaters in Lake Water quality management is entirely different from that in rivers.
- River is a flowing water body while lake has stagnant waters, so in lakes only top surface would become saturated with DO, but the bottom layers would not have enough oxygen.
- Overturning of layers would not occur frequently, so that DO content would not be uniform throughout the depth of lake. Overturning takes place only when there is change in the season due to which there will be temperature difference between water in different layers which cause change in the densities of different layers and overturning occurs.

Factors Affecting Self Purification

- ***Dilution***- discussed earlier
- ***Current*** - When strong water current is available, the discharged wastewater will be thoroughly mixed with stream water preventing the deposition of solids.

In a small current, the solid matter from the wastewater will get deposited at the bed following decomposition and reduction in DO.

- ***Temperature*** - The quantity of DO available in stream water is more in cold temperature than in hot temperature.

Activity of microorganisms is more at the higher temperature, hence, the self-purification will take less time at hot temperature than in winter.

- ***Sunlight*** - Algae produces oxygen in the presence of sunlight due to photosynthesis. Sunlight helps in the purification of streams by adding oxygen through photosynthesis.
- ***Rate of oxidation*** - Due to oxidation of organic matter discharged in the river DO depletion occurs. This rate is faster at higher temperature and low at lower temperature. The rate of oxidation of organic matter depends on the chemical composition of organic matter. 35

Actions involved in Self-purification

- • Dilution – discussed earlier
- • Dispersion due to currents
- • Sedimentation
- • Oxidation
- • Reduction
- • Temperature
- • Sun light

Disposal by Dilution

- *Process in which treated wastewater from ETP is discharged into a large static body of water or in moving water bodies such as rivers or streams*
- *The discharged wastewater is purified in due course of time, by the self purification of natural water*
- *The effluent discharge and degree of treatment of waste water depends upon self purification capacity of the stream and its intended water*

Standards of dilution

Dilution factor	Standards of purification required
Above 500	No treatment required. Raw sewage can be directly discharged into river
Between 300 to 500	Primary treatment such as PST is required so that SS concentration is less than 150 mg/lit
Between 150 to 300	Treatment such as screening, sedimentation and chemical precipitation are required so that SS concentration is less than 50 mg/lit
Less than 150	Thorough treatment is required, SS should be less than 50 mg/lit and BOD 5 should be less than 20 mg/lit

2. Dispersion due to currents

- Self purification largely depends upon currents
- Currents readily disperse the wastewater in the stream, preventing locally high concentrations of pollutants.
- High velocity improves reaeration which reduces the time of recovery (length of stream affected by the wastewater is increased)

3. Sedimentation

- If the stream velocity is lesser than the scour velocity of particles, sedimentation will take place, which will have two effects:
 - *The suspended solids, which contribute largely to oxygen demand, will be removed by settling and hence water quality downstream will be increased.*
 - *Due to settled solids anaerobic decomposition may take place.*

4. Oxidation

- The organic matter is oxidized by aerobic bacteria by utilizing the dissolved oxygen of the natural water.
- The process prevails till complete oxidation of organic matter takes place.
- The stream which is capable of absorbing more oxygen rapidly through reaeration etc. can purify heavily polluted water in a short time.

5. Reduction

- The reduction occurs in the streams due to hydrolysis of the organic matter biologically or chemically.
- Anaerobic bacteria will split the organic matter into liquids and gases, thus paving way for their ultimate stabilization by oxidation.

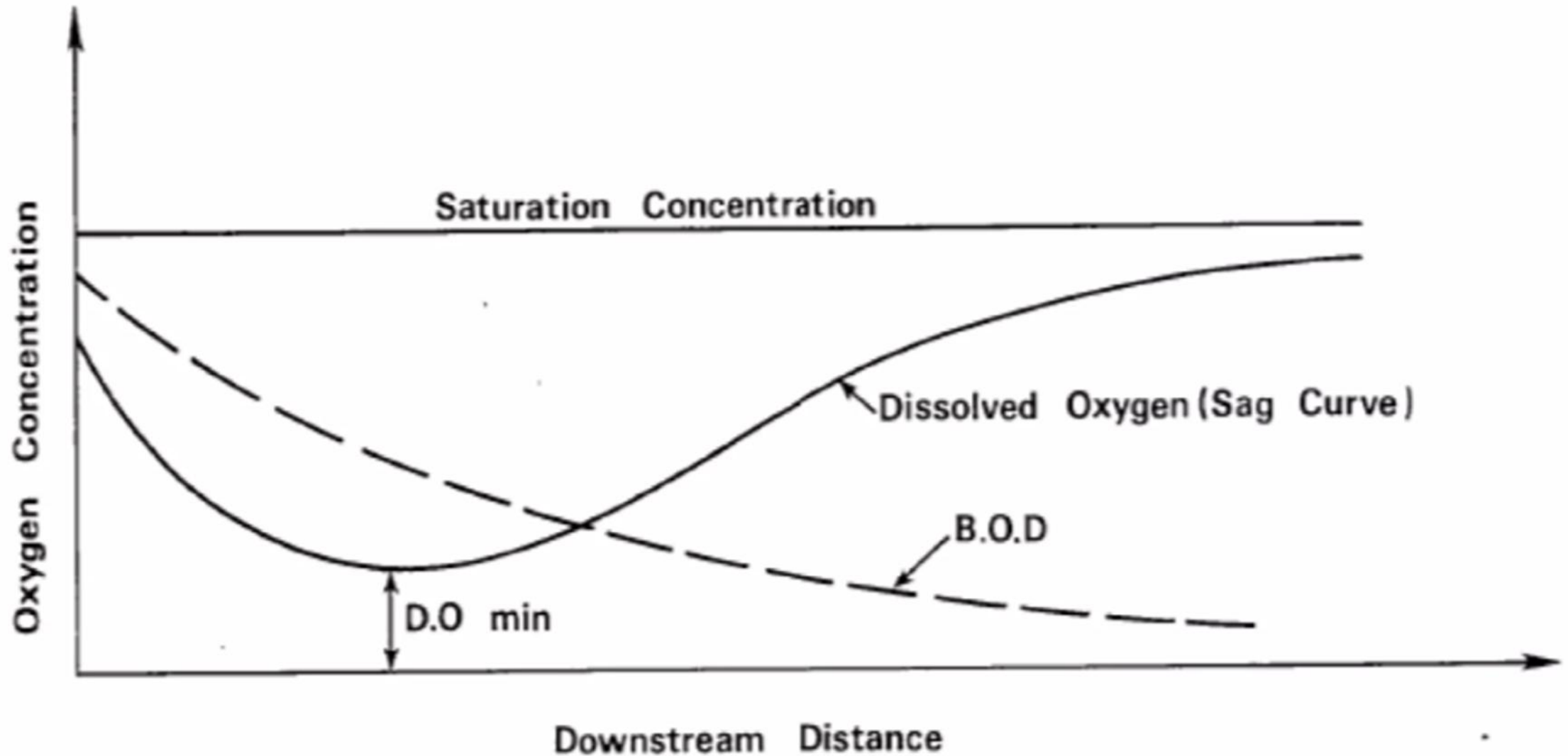
6. Temperature

- At low temperatures, the activities of bacteria is low and hence rate of decomposition will also be slow, though DO will be more because of increased solubility of oxygen in water.
- At higher temperatures, however, the self-purification takes lesser time, though the quantity of DO will be less.

7. Sun Light

- Sun light helps certain microorganisms to absorb carbon dioxide and give out oxygen, thus assisting in self purification.
- Sun light acts as a disinfectant and stimulates the growth of algae which produce oxygen during day light but utilize oxygen at night.
- Whenever there is algal growth, the water may be supersaturated with DO during daylight hours, though anaerobic conditions exist in night.

Development of Oxygen Sag Curve



De- Oxygenation & Re- Oxygenation in River & Streams

- **De- oxygenation**

- When pollution load is discharged in to the Stream/Rivers, DO goes on reducing.
- It depends upon Organic matter present and temperature.
- Variation or depletion of DO is represented graphically by De-Oxygenation Curve.

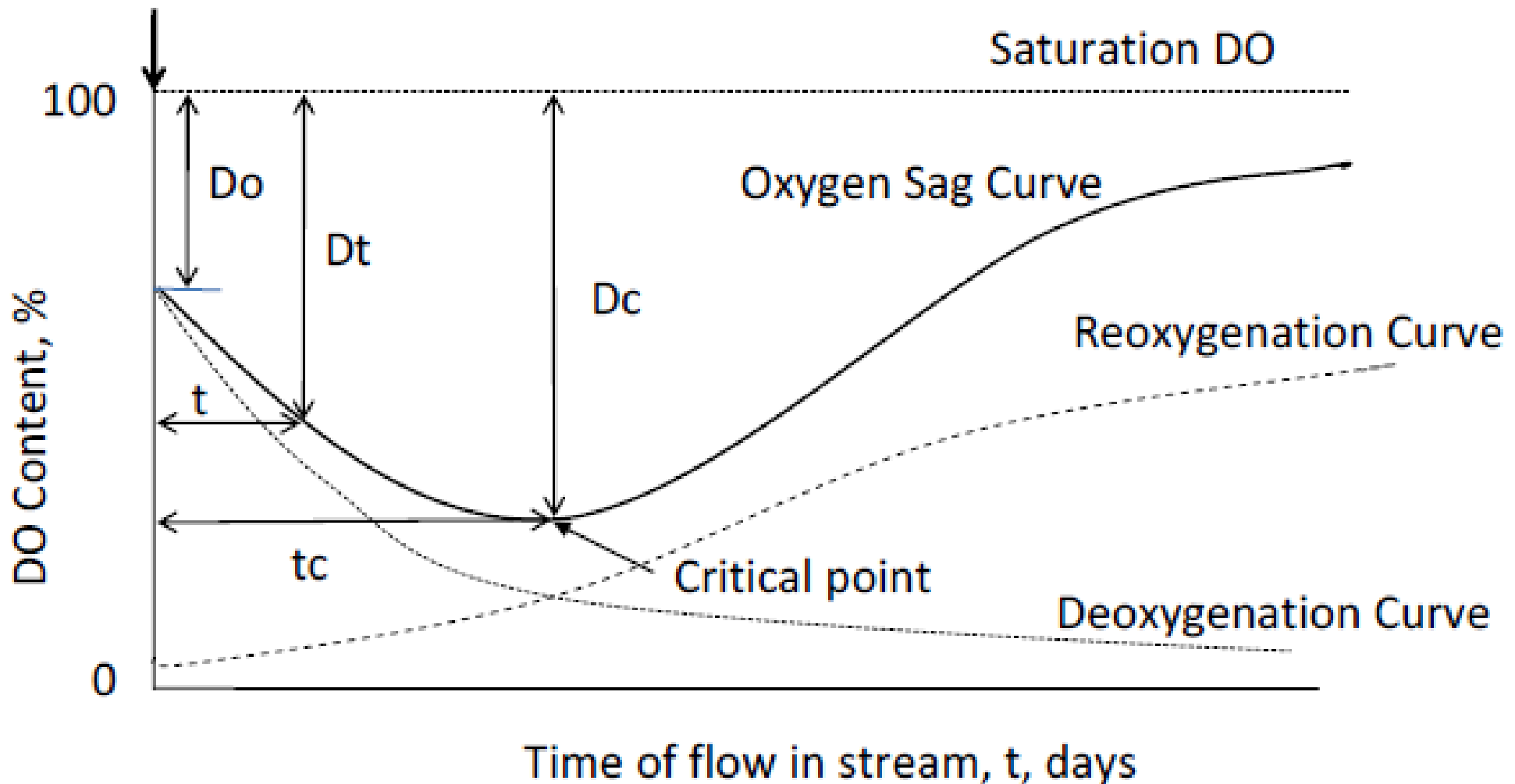
- **Re-oxygenation or Re-aeration Curve**

- At the time of de-oxygenation, oxygen gets added into the stream through various proceses such as photosynthesis, rains etc.
- The curve representing oxygen gaining process Re-oxygenation or Re-aeration Curve

In a running polluted stream de-oxygenation and reoxygenation/reaeration goes hand in hand, If de-oxygenation is more deficit results.

Deoxygenation, reoxygenation and oxygen sag curve

Point of oxygen demanding waste discharge



DO deficit

- The oxygen sag or oxygen deficit in the stream at any point of time during self purification process is the difference between the saturation DO content and actual DO content at that time.
- The amount of DO deficit can be obtained by graphically adding both curves.
- The resultant curve is known as the DO Sag Curve.
- DO deficit is given by

$$\begin{aligned}\text{DO deficit} &= \text{Saturation DO} - \text{Actual DO} \\ &= \text{DO sat} - \text{DO act}\end{aligned}$$

- The saturation DO value for fresh water depends upon the temperature and total dissolved salts present in it; and its value varies from 14.62 mg/L at 0oC to 7.63 mg/L at 30oC, and
- lower DO at higher temperatures.
- Streeter Phelp's equation can be used for analysis of DO Sag Curve.

DO deficit contd....

- The saturation DO value for freshwater depends upon the temperature and total dissolved salts present in it; and its value varies from 14.62 mg/L at 0 C to 7.63 mg/L at 30 C, and lower DO at higher temperatures.
- Streeter Phelps's equation can be used for analysis of DO Sag Curve.
- The oxygen sag curve can be easily done by superimposing the rates of deoxygenation and reoxygenation as suggested by the Streeter – Phelps's analysis

Application of Streeter Phelps's equation

$$dD_t / dt = f \text{ (deoxygenation and reoxygenation)}$$

$$\text{OR } dD_t / dt = K' L_t - R' D_t \quad \dots(1)$$

Where,

D_t = DO deficit at any time t ,

L_t = amount of first stage BOD remaining at any time t

K' = BOD reaction rate constant or deoxygenation constant (to the base e)

R' = Reoxygenation constant (to the base e)

t = time (in days)

dD_t / dt = rate of change of DO deficit

Application of Streeter Phelp's equation Contd....

Changing base of natural log to 10 the equation can be expressed as:

$$Dt = \frac{KLo}{R - K} [10^{-K.t} - 10^{-R.t}] + Do.10^{-R.t} \quad \dots(5)$$

Where, K = BOD reaction rate constant, to the base 10

R = Reoxygenation constant to the base 10

Do = Initial oxygen deficit at the point of waste discharge at time t = 0

t = time of travel in the stream from the point of discharge = x/u

x = distance along the stream

u = stream velocity

The rate of reoxygenation depends upon:

- i) Depth of water in the stream: more for shallow depth.
- ii) Velocity of flow in the stream: less for stagnant water.
- iii) Oxygen deficit below saturation DO: since solubility rate depends on difference between saturation concentration and existing concentration of DO.
- iv) Temperature of water: solubility of oxygen is lower at higher temperature and also saturation concentration is less at higher temperature.

Disposal to Lakes

- **Lake Pollutants**
- In Lake the phosphorous a nutrient largely contained in industrial domestic wastewater is seriously affecting the quality of lakes and hence it is considered as the prime lake pollutant.
- Oxygen demanding wastes may be the other important lake pollutant. The toxic chemicals from industrial wastewater can be present.

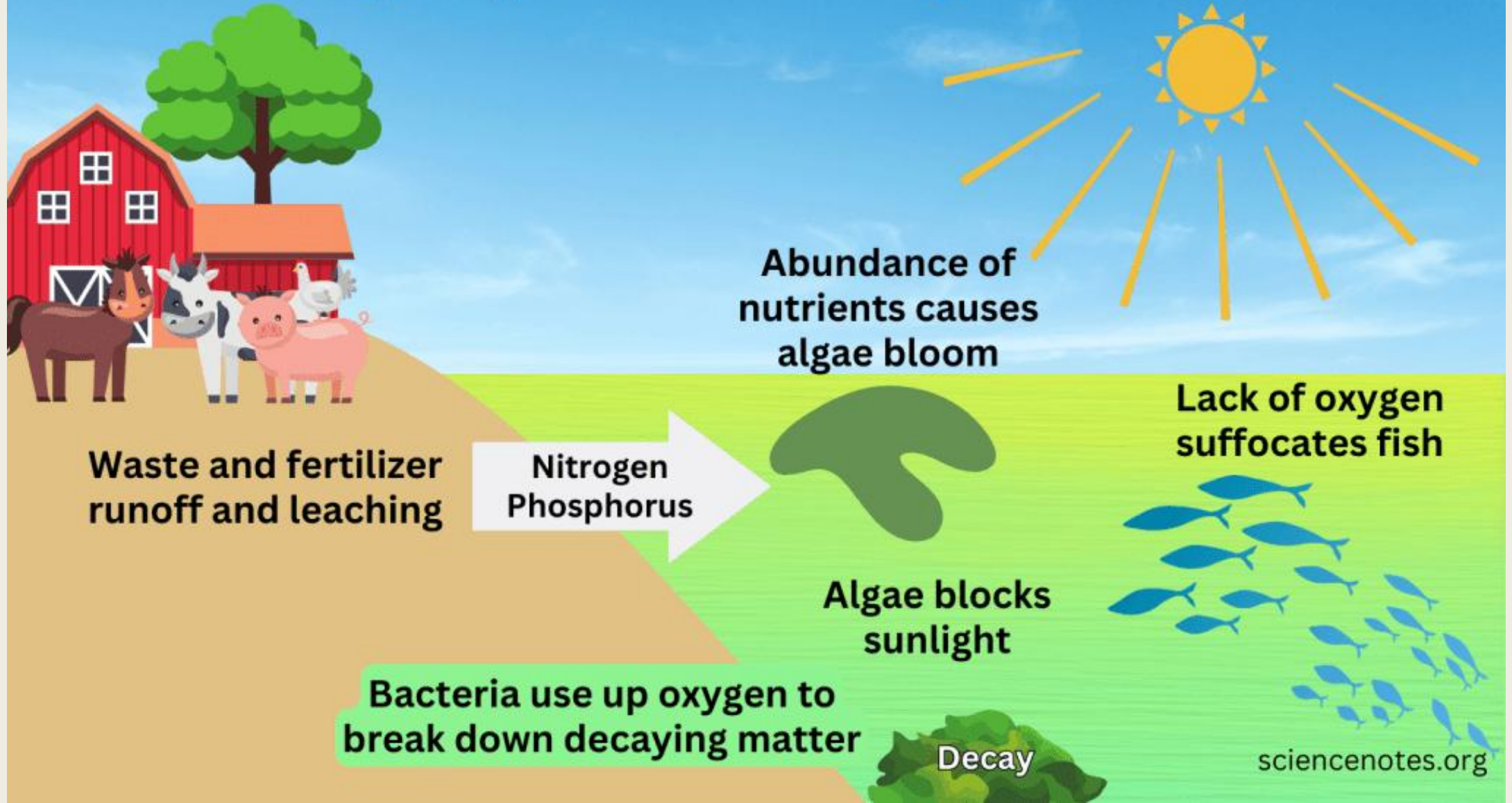


Eutrophication

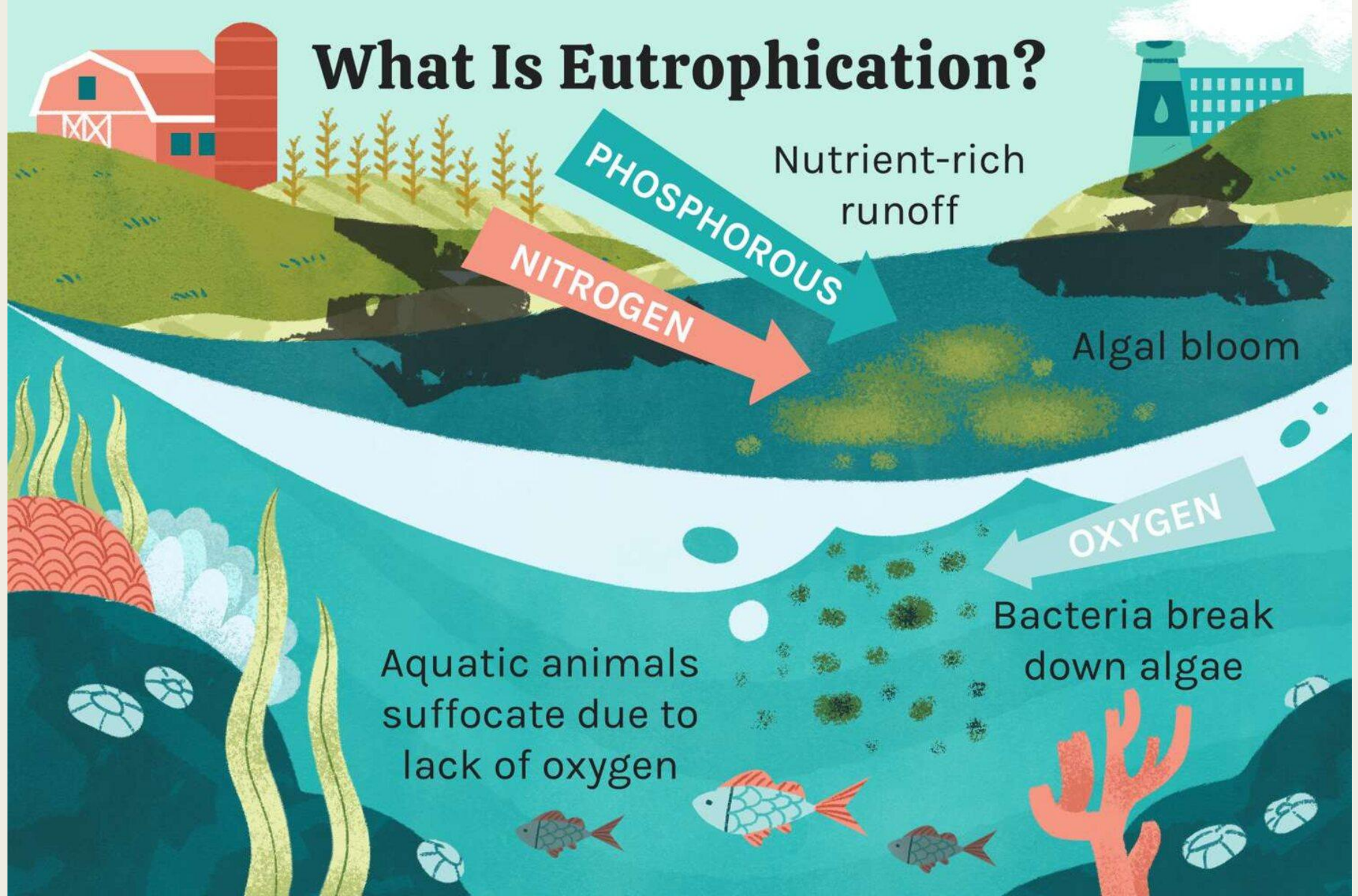
- Phosphorous acts as the nutrients for the algal growth.
- Increase in the phosphorous content would increase the algal growth.
- Excessive in the phosphorous content would increase the algal growth.
- Excessive algal growth (Algal Bloom) -----lot of the problems like taste, odour, problems in oxygen diffusion in lower layer.
- Ultimately the entire lake can get covered with algae and it may become useless for other organisms

What Is Eutrophication?

Eutrophication is the excessive nutrient enrichment of water, leading to algal blooms and ecological imbalance.



What Is Eutrophication?



Disposal to Sea or Ocean

- The saturation concentration of dissolved oxygen in water decreases with increasing salt concentration.
- DO in sea water is approximately 80 % less than that in water.
- In addition to this deficiency, the temperature of sea water is lower than the sewage temperature, whereas the specific gravity is higher.
- Due to this reasons, when the sewage is discharged into the sea water, the lighter and warmer sewage will rise up to the surface, resulting in spreading of the sewage at the top surface of sea in a thin film or 'sleek'





- Moreover seawater contains a large amount of dissolved matter which chemically reacts with the sewage solids, resulting in the precipitation of some of the sewage solids, giving a milky appearance to the seawater
- Resulting in the formation of sludge banks and a thin milky layer formed at the top of seawater producing offensive hydrogen sulphide gas by reacting with sulphate rich seawater
- As such the capacity of seawater to absorb sewage solids is not as high as that of fresh water of a river.
- DO content is less.
- However, since the sea contains a large volume of water, most of these deficiencies can be overcome if the sewage is discharged deep into the sea, much away from the coastline, with extreme care.

Points be kept in mind while discharging sewage into the sea

- The sewage should be discharged in deep sea water.
- In order to mix sewage properly with the seawater, the sewage should be released at a **minimum depth of 3 to 5 m below the water level** and taken sufficiently inside the shore, thus preventing nuisance to baths and recreation centres on the shore.
- To prevent the backing up and spreading of sewage on the sea shore, the sewage should be disposed of only during low tides, large sized tanks may, therefore be constructed to hold the sewage during high tides. The provision of a large-sized sewer, grated with a non-return valve at the end, is also an alternative to hold the sewage during high tides.

Points be kept in mind while discharging sewage into the sea contd...

- Before deciding the position of the outfall point, the sea current, wind directions, velocity, etc., should be properly studied. The sea currents, wind direction, velocity, etc. should be properly studied. The point of disposal should be such that the sewage is taken away from the shore by the winds, and not brought back near the shore by the winds, and not brought back near the shore.
- The outfall sewer should be placed on a firm rocky foundation, and encased in thick stone masonry, so as to properly protect it from wave action, floating debris, etc..
- The discharge of Industrial waste waters into sea should however, be controlled in respect of the quality of the effluents, by adhering to the following Indian standards., prescribed by IS 1968-1976.

Comparison of Disposal Methods

- For most of Indian towns and cities land disposal is a better choice for the following reasons:
- India is a tropical country and temperature remains high most of the year. Due to the hot climate, the DO content of the river water reduces, while the fish and aquatic life require higher DO for their survival, thus making the dilution method more difficult and prohibitive.
- Most of the rivers have a very small amount of dry weather flow (in summer) so the amount of dilution available would be less. Also in most of the cities these rivers are the only source of drinking water. So disposal by dilution method cannot be used, and if used, a very high degree of treatment is required which would increase the cost of disposal.

Comparison of Disposal Methods contd...

- Except for a few major cities, the water supplies in India are very low, thus resulting in the production of highly concentrated sewage, which on travelling in hot climate becomes stale and septic by the time it reaches the disposal point, thus prohibiting the use of dilution method for disposal.
- There are only a few coastal towns in India, which have strong tidal currents moving in the forward directions, and the necessary depth of water at the point of disposal, thus prohibiting the disposal of sewage in the sea on large scale.

Water Reclamation and Reuse

- **Recirculation of Industrial Wastewater**
- Recirculation is circulating something again for reuse, or recycling to minimize process cost and increase efficiency.
- Eg: recirculating water is a way of reducing the load of organic waste in the water and of increasing the amount of oxygen in the water.
- Effluent is treated in a wastewater treatment plant through recirculation, where it is piped back to the beginning of the treatment process and added to the raw wastewater.
- A recirculation pump can be used in a system to ensure that hot water is always available as close to the consumption point as possible, in order to reduce water wastage and to increase comfort.

Water Reclamation and Reuse contd...

- **Recirculation of Industrial Wastewater**
- A recirculation system can save thousands of gallons of fresh water annually.
- All cooling towers offer the potential for recirculation, the extent of which depends primarily upon the entering and exiting air velocities, and their relationship to each other.
- Higher entering velocities increase the potential for recirculation, while higher exit velocities decrease its opportunity.

Water Reclamation and Reuse contd...

- **Recirculation of Industrial Wastewater**
- Recirculation is very important in many systems where water recirculates through the cooling pipes for absorbing heat so that system can run uninterrupted, like:
 - Air conditioners
 - Refrigerators
 - Heat exchangers
- Recirculation is very common in industrial and mining applications. In mining industries, almost all of the recalculated water (98%) is used for process activities. In thermal-electric power, the recirculation rate is 16%.



Industrial Water Recycle & Reuse

- Industrial water reuse and recycling is the process by which wastewater produced from one source is treated to be reused in the same process or recycled for another.
- Various methods for recycling or reusing industrial water are available, depending on water quality requirements, space constraints, and budgetary considerations.
- Benefits can include the reduction of freshwater costs, wastewater flows, and the size of your water footprint.
- Operational efficiency and sustainability can also be increased along with improved production capacity due to the increase in available clean water.

Industrial Water Recycle & Reuse Contd...

- Reducing the amount of water use by industries can lower water withdrawals from local water sources thus increasing water availability and improving community relations, increasing productivity per water input, lowering waste water discharges and their pollutant load, reducing thermal energy consumption and potentially processing cost.
- Wastewater reuse in industry can take place within a business or between businesses and has the potential to reduce costs for businesses both on water bills as well as wastewater treatment.
- Depending on the contaminants present in wastewater and its future reuse, it can either be directly reused, or treated and reused (recycled).

Advantages

- Reduces the amount of water used.
- Reduces water bills.
- Reduces the volume of generated wastewater (no waste).
- Reduces costs through industrial symbiosis (by-product reuse, sharing management of utilities, sharing ancillary services).

Disadvantages

- Requires high knowledge about quality of water for reuse.
- Requires financial investments.
- Requires a high level of trust between industries.
- Requires modification of current operations both for direct reuse and treat-and-reuse.

Use of Municipal Waste Water in Industries

- As an alternative water resource, wastewater reuse is widely known to have many benefits.
- However, a successful application depends on a number of factors, of which quantity and quality must receive great attention.
- The former deals with how reliable and dependable the wastewater source is.
- The latter concerns how safe the reused water would be to human beings and the relevant ecosystem, in particular for long-term application.

Use of Municipal Waste Water in Industries contd...

- **Municipal wastewater reuse** -----classified as **direct and indirect reuse**.
- **Direct reuse** requires distribution systems to deliver treated wastewater to the users.
- It is planned, Deliberate or intentional use of treated wastewater for some beneficial purposes.
- Direct reuse of treated wastewater for Drinking water, however, is not a viable option due to health risk concerns.
- **Indirect reuse** refers to the use of treated wastewater after its return to natural water sources (i.e. river, lake and aquifer) for purification and dilution.
- It involves natural buffers for greater temporal and spatial separation of treatment.

- Though being strongly associated with locality, the following types of reuse applications have been applied to treated wastewater
- • Agricultural irrigation;
- • Industrial reuse;
- • Groundwater recharge;
- • Urban reuse;
- • Augmentation of potable supplies; and
- • Recreational and habitat restoration/enhancement.

The potential for any of these applications is subject to the control of pathogens, biodegradable and refractory pollutants, heavy metals and other toxic organics.

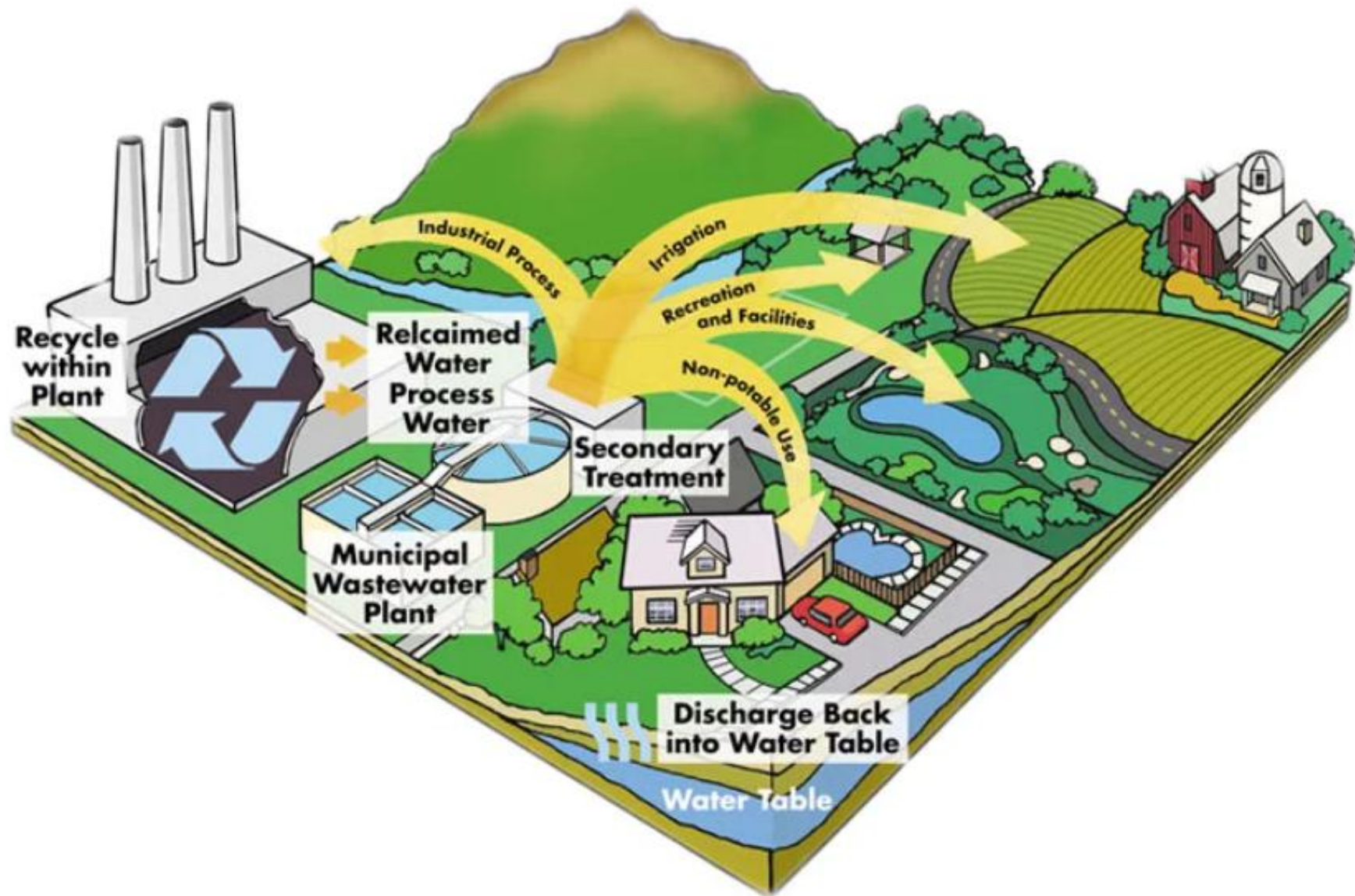
Since municipal wastewater treatment plants may receive toxic chemicals from industries, effective pretreatment and pollution prevention programs are absolutely necessary prior to the consideration of any potential reuse application

Industrial reuse:

- Cooling water
- Boiler feed
- Process water
- Heavy Construction
- Ion exchange rinse water

Depending on the specific reuse application, a high level of treatment provided by the municipal wastewater treatment facilities may be required in order to protect human health and the environment.

These concerns thus necessitate the formation of criteria, standards and guidelines that are appropriate for the users or consumers of this water, which again is manifested via its physical, chemical, biological and even radiological characteristics



Assignment Questions from Module II

- Identify the applications of Streeter Phelp's equation and write the equation
- Write short notes on
 - *Significance of oxygen sag curve*
 - *Waste water disposal in (i) lakes and (ii) ocean*
- List the various methods of wastewater reclamation and reuse
- Express the importance of re-oxygenation and de-oxygenation in problems of stream sanitation